Sprint 4 Review Date: Thursday April 30th
Scrum Master for Sprint 4: Varun Hari

Section 0 – Sprint 3 Reality Check (Required)
You must use evidence from your Sprint 3 demo, GitHub activity, or board behavior. Vague answers will not receive full credit.

- What specifically broke during your Sprint 3 demo?
  Several products weren't in the Open Food Facts database, which meant the app returned incomplete or generic information for those scans, making the AI breakdown less useful for those cases.
- Which ONE issue is most likely to happen again in Sprint 4?
  Board not updated
- What evidence shows this is still a real risk?
  (Must reference demo behavior, GitHub activity, or board patterns)
  In Sprint 3, PRs were being opened but not consistently linked to their corresponding board tasks, which means the board didn't always give an accurate picture of what work was tied to what feature. This pattern is likely to continue unless we enforce it actively.

Section 1 – Retrospective Rule Application

- Copy your rule for Sprint 4 from the Sprint 3 Change Plan (exact wording):
  At the end of each class period, Varun Hari will make sure that the entire GitHub board is updated and each PR (if made that class period) is tied to a certain task
- When does this rule happen during a normal class period?
  (Be specific: start, during work time, end of class, etc.)
  At the end of each class period during the last 5 minutes before dismissal
- What would I SEE happening if this rule is followed correctly?
  Varun opens the GitHub project board and repository, checks every open or recently merged PR for a linked issue, and moves any unlinked PR back to in progress until the author links it
- Where will this show up in your workflow? (select all that apply)
  ☐ Task creation
  ☒ Task completion
  ☐ Testing
  ☒ Board updates
  ☒ Pull requests
  ☒ Daily Scrum
- What is ONE way this rule could still fail in Sprint 4?
  A team member opens a PR and merges it quickly before Varun does the end of class check, bypassing the linking requirement

- What happens immediately if this rule is NOT followed?
  The PR is blocked from being reviewed or merged and the corresponding board task is moved back to in progress until the issue is linked

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Section 2 – Sprint Goal (Expansion + Stability)
Your goal must add new value WITHOUT breaking your existing system.

- Sprint Goal (By the end of Sprint 4, a user will be able to…):
  By the end of Sprint 4, users will be able to receive AI-powered recommendations for 1-3 healthier substitute products based on their scanned item and view a direct nutritional comparison between two products.
- What existing system MUST remain stable?
  The core AI analysis pipeline for the initially scanned product has to remain stable and continue returning baseline results in under 10 seconds.
- What would failure look like at demo?
  The AI hallucinates a substitute product that does not exist, or the app times out and crashes while trying to process data for multiple products simultaneously

Section 3 – Scope Control (Hard Constraint)

- What IS included in this Sprint (2 – 4 items MAX):

Backlog Item	**Why it supports the goal**
See healthier substituted product recommendations	This fulfills the core sprint goal by querying the database for alternatives in the same category
Comparison of nutrient metrics between products	Provides the user with the direct AI-driven comparison necessary to understand why the substitute is better
Click or tap here to enter text.	Click or tap here to enter text.
Click or tap here to enter text.	Click or tap here to enter text.

- What is NOT included in this Sprint (REQUIRED)
  We are not building out social media sharing features, daily nutrition diaries, or account login systems
- What are you intentionally delaying and why?
  We are intentionally delaying the manual text search fallback from our previous ideas since implementing a multi-product AI comparison is highly complex and token-heavy, adding text search at the same time in the same sprint risks breaking the scope constraint of a 10-second response time
- Why is this amount of work realistic for your team this Sprint?
  We are strictly focusing on data we already know how to fetch. We are simply

expanding the OpenAI prompt to analyze multiple JSON objects (the scanned item + the database substitutes) instead of just one.

Section 4 – Scope Validation
Do NOT list new backlog items. Use only the items from section 3.

- Which item is the highest risk to your Sprint Goal? Why?
  The comparison of nutrient metrics. Feeding multiple product ingredient lists into the OpenAI API simultaneously risks increasing latency and exceeding token limits
- Which item improves stability or user experience?
  The healthier substitute recommendations improve the UX by giving the user an actionable next step, rather than just telling them their current food is unhealthy
- If you had to remove ONE item by Dev Day 3, which would it be? Why?
  The direct side-by-side UI comparison of two products, we would fall back to just having the AI output a text paragraph suggesting the substitute
- Why is this total scope realistic for your team?
  Our team has already successfully managed complex asynchronous API calls (OpenAI) and Azure database routing and these additions build directly on the parsing logic we established in Sprints 1 and 2

Section 5 – Task System (Strict Execution)
Tasks MUST be ≤ 1 class period.
If the definition of done is vague, the task is invalid.

Add additional rows to this table as necessary by copy and pasting the contents of an existing row and then modifying.

Task	Owner	Backlog Item	Done = (clear criteria)	Includes Testing?	How is it tested?
Query Open Food Facts for similar items	Siddharth	Substitute recommendations	The app successfully fetches 1-3 items from the same category as the scanned product.	Yes	Console printout.
Refine OpenAI prompt for multi-product comparison	Arda	Comparison of metrics	AI successfully returns a formatted comparison of the scanned	Yes	API test and search results coming

			item vs. substitutes without timing out.		up <10 seconds.
Build UI cards for substitute recommendations	Varun	Substitute recommendations	The UI displays the name, brand, and image of the recommended alternatives below the main scan.	Yes	Visual check on Android emulator.
Build UI for side-by-side metric comparison	Leo	Comparison of metrics	Tapping a substitute expands a view showing macro/ingredient differences clearly.	Yes	End-to-end tap test on device.
Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Click or tap here to enter text.	Choose an item.	Click or tap here to enter text.

Section 6 – Testing Integration

- When during the task is testing performed?
  Testing is performed locally by the developer immediately after code is written, and then verified again by the peer reviewer before the PR is approved.
- Who verifies the task meets Definition of Done BEFORE it moves to Done?
  The assigned PR Reviewer (a team member who did not write the code).
- What happens immediately if a test fails?
  The PR is rejected with comments, the code remains on the feature branch, and the board task stays in "In Progress."

Section 7 – Workflow Discipline

- When EXACTLY is the board updated each class?
  (Example: last 3 minutes of class before dismissal)
  Within the first 5-10 minutes of our 50-minute class periods
- Who checks that the board reflects reality?
  Varun (Scrum Master)

- How do tasks move (To Do -> In Progress -> Done)?
  No new tasks can be moved to In Progress within 48 hours of the Sprint Review. All Pull Requests must be submitted at least 24 hours before the review date.
- Rule to prevent last-minute work completion:
  No new tasks can be moved to In Progress within 48 hours of the Sprint Review. All Pull Requests must be submitted at least 24 hours before the review date

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Section 8 – Failure Prevention Plan

- Early warning sign that your Sprint is failing:
  The OpenAI API takes longer than 15 seconds to return the comparison data, causing the UI to freeze or the user experience to degrade
- Trigger condition (REQUIRED FORMAT):
  If [X] is not complete by [time], we will [action]
  If the AI comparison latency is consistently over 10 seconds by Dev Day 5, we will remove the real-time AI comparison generation and only display the raw nutritional data for the substitute products
- Who is responsible for making this adjustment?
  Varun (Scrum Master) and Siddharth (API Lead)
- What will you REMOVE from scope if needed?
  The side-by-side nutrition comparison generated by AI

Section 9 – Expansion Justification

- Why is this the right feature to build now?
  It shifts the app from being just an information delivery tool to an actual health improvement tool, which aligns with our primary persona of a user wanting to make better health choices
- How does it improve user experience or product value?
  It prevents the user from feeling helpless if they scan a bad product by immediately offering them a healthier, vetted alternative option that they can take
- Why is your system ready to handle this expansion?
  (Must reference a specific system improvement from your Sprint 3 retrospective)
  In Sprint 3, we successfully stabilized the AI processing speeds to under 10 seconds. Because the baseline API pipeline is now refined, we have the ability to handle slightly larger payloads without breaking the app.

Section 10 – Final Commitment Checklist

☒ Our rule is observable and enforceable
☒ All tasks meet the ≤1 class period requirement.
☒ Every task includes testing
☒Scope is realistic and constrained

☒Board update expectations are clear
☒We have a plan to detect failure early